

Summary: Normalizing Flows for Spectrum Analysis

GitHub repository: <https://github.com/jlvi179/Advanced-Deep-Learning>

What the Notebook Does

This notebook trains models using Conditional Normalizing Flows (via the `jammy_flows` library) to estimate stellar parameters (Effective Temperature T_{eff} , Surface Gravity $\log g$, and Metallicity $[\text{Fe}/\text{H}]$) directly from spectra. Like the uncertainty model, it goes beyond point estimation to output full Probability Density Functions (PDFs) by minimizing the Negative Log-Likelihood (NLL).

The notebook evaluates three specific normalizing flow configurations appended to a Convolutional Neural Network feature extractor: a Diagonal Gaussian, a Full Gaussian, and a Full Flow model. The evaluation process highlights how well the models can capture both independent uncertainties and complex physical correlations among the target labels, culminating in multi-dimensional PDF visualizations and calibration metric comparisons.

Differences Between the Models

The core differences lie in their probabilistic output layers and flexibility:

- **Diagonal Gaussian (`diagonal_gaussian`):** Outputs independent errors. It cannot model associations or correlations between parameters (its 2D contours are always axis-aligned).
- **Full Gaussian (`full_gaussian`):** Predicts a full covariance matrix. This allows it to model linear dependencies and tilts (correlations) between variables like temperature and surface gravity.
- **Full Flow (`full_flow`):** Implements a deep, non-linear flow architecture. It can learn highly complex, skewed, or multimodal distribution shapes, making it the most robust for handling physical degeneracies in spectral measurements.

Fine points to watch:

- **PDFs over Point Estimates:** Normalizing flows require sampling the learned distribution to extract mean target values and standard deviations.
- **Correlation Modeling:** The 3x3 evaluation plots illustrate marginal distributions on the diagonal and the pairwise correlations in the off-diagonal hexbin plots.
- **Calibration (Pull Statistics):** Just like in standard uncertainty prediction, evaluating whether the error matches the predicted standard deviation ($\text{Pull} \approx 1.0$) determines if the model is over- or under-confident.
- **Capacity vs. Stability:** While the full flow provides maximum flexibility, deeper flow models require precision control (e.g., converting to `float64`) to ensure numerical stability during the log-likelihood evaluation.

In short, replacing standard output heads with Normalizing Flows transforms the network into a powerful density estimator capable of expressing physical correlations and complex error contours rather than simply symmetric Gaussian uncertainty.